

DISCoRE-IQA: DISENTANGLED CONTENT-QUALITY REPRESENTATION AND REGION-LEVEL SEGMENTATION FOR LIGHTWEIGHT IMAGE QUALITY ASSESSMENT

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ABSTRACT

Lightweight text-guided Image Quality Assessment (IQA) frameworks bypass multi-billion parameter Multimodal Large Language Model (MLLM) inference costs by distilling reasoning trajectories into visual prototype transport heads. However, existing distilled architectures suffer from severe semantic-quality feature interference and fail to detect spatially localized visual defects. In this paper, we introduce **DisCoRe-IQA (Disentangled Content-quality Representation + Region-level quality segmentation)**, a lightweight, reasoning-free framework for robust cross-domain visual quality scoring. DisCoRe-IQA incorporates three core technical innovations: (1) Stage-1 Low-Rank Adaptation (LoRA) of a vision-language backbone; (2) Supervised Adversarial Content-Quality Disentanglement using a Gradient Reversal Layer (GRL) trained against semantic target clusters to explicitly unbind content bias from the quality representation space; and (3) a Region-Level Quality Segmentation Head trained on synthetic localized defects and real Mean Opinion Score (MOS) Multiple Instance Learning (MIL) anchoring. Diagnostic content-probe classification accuracy drops from 0.868 in the raw feature space to 0.138 in our disentangled space, confirming genuine content invariance. Furthermore, our patch segmentation head achieves an 89.6% worst-region defect localization hit-rate on held-out localized distortions. Extensive evaluations across seven benchmark datasets (*KonIQ-10k*, *SPAQ*, *KADID-10k*, *PIPAL*, *LIVE-Wild*, *AGIQA-3K*, *CSIQ*) demonstrate that DisCoRe-IQA achieves state-of-the-art performance (+0.0062 average PLCC, +0.0084 average SRCC over the RALI baseline) while executing in a single vision-encoder forward pass (< 3 ms latency, zero LLM FLOPs at inference).

1 INTRODUCTION

No-Reference Image Quality Assessment (NR-IQA) is a fundamental capability in visual perception systems, image restoration pipelines, and digital media processing Su et al. (2020); Ke et al. (2021); Tian et al. (2025). Traditional deep learning approaches primarily map visual features directly to scalar Mean Opinion Scores (MOS) Yang et al. (2022); Zhang et al. (2023). Recently, the emergence of Multimodal Large Language Models (MLLMs) and vision-language foundation backbones Jia et al. (2021); Radford et al. (2021); Wang et al. (2022) has catalyzed a paradigm shift toward language-guided visual scoring Wu et al. (2024b; 2023b); Li et al. (2025). Advanced reasoning-based MLLMs, such as Q-Insight Li et al. (2025), utilize reinforcement learning (e.g., Group Relative Policy Optimization) to generate explicit reasoning chains (`<think>...</think>`) prior to outputting quality scores. By converting high-dimensional visual inputs into compact, cross-domain aligned textual representations, these reasoning models achieve exceptional out-of-domain generalization across diverse image distributions.

Despite their strong generalizability, reasoning-induced MLLMs impose prohibitive computational costs Zhao et al. (2025); Liu et al. (2024); Fan et al. (2026). Generating textual reasoning tokens through multi-billion parameter autoregressive decoders consumes substantial GPU memory and introduces multi-second inference latency, restricting their deployment in real-time or resource-

constrained applications. To circumvent these computational bottlenecks, lightweight text-aligned architectures such as RALI Zhao et al. (2025) distill MLLM reasoning outputs into a vision-encoder feature space. RALI aligns image embeddings directly with quality reasoning prototypes using parameter-efficient fine-tuning (PEFT) Chen et al. (2022); Yin et al. (2023); Jia et al. (2022); Jie & Deng (2022) and bucketed k -means prototype transport Chen et al. (2018); Rymarczyk et al. (2021), enabling single-forward-pass scoring (< 3 ms latency, zero LLM generation FLOPs).

However, current prototype-based distilled IQA frameworks suffer from two major architectural limitations. First, they exhibit severe **semantic-quality representation entanglement** Yang et al. (2025); Liu et al. (2025). Pre-trained vision encoders (e.g., CLIP-ViT) are fundamentally optimized for high-level semantic alignment. Consequently, prototype vectors constructed in raw or unsupervised PCA-nulled visual feature spaces remain dominated by semantic content (e.g., object categories like “cat” vs. “building”) rather than pure perceptual degradation signals. Cosine similarities to quality prototypes are thus easily corrupted by high-level visual semantic similarity, causing false quality predictions when visual content interferes with degradation severity.

Second, prototype transport heads trained strictly under global scalar MOS supervision lack explicit localized defect modeling and region-aware spatial reasoning Ke et al. (2021); Su et al. (2020). When an image suffers from localized severe artifacts (such as spatially bounded motion blur, local compression, or blockiness), global pooling operations dilute these critical spatial extrema into a moderate overall score, leading to failures on spatially localized distortions across out-of-domain benchmark datasets Hosu et al. (2019); Fang et al. (2020); Lin et al. (2019); Gu et al. (2020); Ghadiyaram & Bovik (2015); Li et al. (2023); Ponomarenko et al. (2015).

To resolve these fundamental representational bottlenecks, we propose **DisCoRe-IQA** (**Dis**entangled **C**ontent-quality **R**epresentation + **R**egion-level quality **s**egmentation), a lightweight, reasoning-free IQA framework designed for robust cross-domain generalization and localized quality assessment. DisCoRe-IQA introduces two primary technical contributions built on top of a LoRA-adapted vision backbone:

1. **Supervised Adversarial Content-Quality Disentanglement:** Rather than unsupervised dimensionality reduction, we introduce an explicit Gradient Reversal Layer (GRL) content classifier trained against frozen semantic target clusters, actively unbinding semantic category information from the quality embedding while a metric-weighted prototype transport head preserves quality relevance. Linear content-probe accuracy drops from 0.868 (raw features) to 0.138 in our disentangled space, and collapses toward chance on independent *ImageNet/COCO* labels (§4.6).
2. **Region-Level Quality Segmentation:** We reformulate patch-level quality modeling as region-level segmentation, training a Patch Quality Head with joint synthetic localized distortions (exact per-patch severity) and real-MOS Multiple Instance Learning (MIL) anchoring. On a held-out benchmark it reaches strong per-patch correlation ($r = 0.530$) and an 89.6% worst-region hit-rate, matching a grounding MLLM at $\sim 2,949\times$ lower latency (§4.6).

Extensive evaluations across seven benchmarks (*KonIQ-10k*, *SPAQ*, *KADID-10k*, *PIPAL*, *LIVE-Wild*, *AGIQA-3K*, *CSIQ*) plus out-of-domain *TID2013* show DisCoRe-IQA consistently improves over the lightweight RALI baseline (average PLCC 0.805 vs. 0.798, SRCC 0.787 vs. 0.779) in a single forward pass, while additionally equipping lightweight assessors with verified content invariance, spatial defect localization, and robust zero-shot transfer to AI-generated and low-level-perception benchmarks.

2 RELATED WORK

2.1 MULTIMODAL LLMs FOR IMAGE QUALITY ASSESSMENT

The evolution of Multimodal Large Language Models (MLLMs) has transformed visual quality assessment from static score regression into descriptive, multi-dimensional visual understanding Wu et al. (2024b); You et al. (2023; 2024). Supervised fine-tuning (SFT) methods, such as Q-Align Wu et al. (2023b) and DeQA-Score You et al. (2025), model continuous human rating distributions via discrete text-defined levels and soft distribution targets, facilitating multi-dataset co-training while

reducing perceptual scale mismatch. To address low-level perceptual attributes, Q-Instruct Wu et al. (2023a) introduces large-scale instruction tuning specifically tailored for low-level visual feedback.

Recent research integrates Reinforcement Learning (e.g., GRPO) to incentivize explicit step-by-step visual reasoning: Q-Insight Li et al. (2025) and VisualQuality-R1 Wu et al. (2025) generate structured reasoning text prior to scoring, while agentic/memory frameworks such as Q-Probe Li et al. (2026) and ME-IQA Fan et al. (2026) explore test-time probing and re-ranking. Grounding-oriented MLLMs, including Grounding-IQA Chen et al. (2024c) and Q-Ground Chen et al. (2024a), add bounding-box localization and distortion segmentation. However, MLLM assessors suffer substantial latency, high memory, and autoregressive generation bottlenecks. DisCoRe-IQA matches or exceeds MLLM cross-domain accuracy in a single vision-encoder forward pass (< 3 ms, zero LLM generation FLOPs).

2.2 LIGHTWEIGHT DISTILLED AND TEXT-ALIGNED IQA FRAMEWORKS

To bypass MLLM inference overhead, lightweight IQA architectures align visual embeddings with language priors or quality prototypes Wang et al. (2022); Zhang et al. (2023); Zhao et al. (2025). Self-supervised and contrastive frameworks such as CONTRIQUE Madhusudana et al. (2021), Re-IQA Saha et al. (2023), ARNIQA Agnolucci et al. (2024), DCL Bi et al. (2026), and SLIDE-IQA Durbha et al. (2026) learn quality-sensitive latent spaces via degradation classification or spatial contrastive objectives, while prompt-guided PromptIQA Chen et al. (2024b), causal models Tian et al. (2024), and top-down semantic guidance Chen et al. (2023) attempt to isolate content from quality.

Recently, RALI Zhao et al. (2025) introduced a reasoning-aligned lightweight scheme that fine-tunes a vision encoder on offline MLLM reasoning outputs via contrastive learning Radford et al. (2021) and maps visual features to basic quality prototypes via PCA dimensionality reduction and bucketed k -means clustering. Nevertheless, prototype vectors constructed in raw or unsupervised PCA-nulled visual feature spaces remain heavily dominated by high-level semantic categories, leading to corrupted prototype transport when visual content interferes with quality degradation. DisCoRe-IQA addresses this limitation by incorporating explicit, supervised adversarial content removal using a Gradient Reversal Layer (GRL) trained on frozen semantic target clusters.

2.3 LOCALIZED DISTORTION AND REGION-AWARE QUALITY ASSESSMENT

Capturing spatially localized defects remains a core NR-IQA challenge Su et al. (2020); Ke et al. (2021); Yang et al. (2022). Early patch-based CNNs used weighted average pooling of local estimates Bosse et al. (2016), while transformer models such as MUSIQ Ke et al. (2021), MANIQA Yang et al. (2022), and TRIQ Golestaneh et al. (2021) model non-local patch interactions via multi-scale and channel-spatial attention; recent works add peak-guided calibration Zhou et al. (2026), compensation restoration Pan et al. (2022), and meta-learning priors Zhu et al. (2020). However, patch heads trained strictly under global MOS supervision struggle to learn precise distortion boundaries from weak scalar feedback. DisCoRe-IQA instead reformulates patch quality as region-level segmentation trained on explicit synthetic localized targets plus real-MOS MIL anchoring, enabling reliable spatial defect localization alongside robust global scoring.

3 METHODOLOGY

3.1 FRAMEWORK OVERVIEW & STAGE-1 BACKBONE ADAPTATION

As illustrated in Figure 1, DisCoRe-IQA processes an input image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ using a pre-trained CLIP-ViT-L/14-336 vision encoder Radford et al. (2021). The encoder extracts a global class token $\mathbf{z}_{\text{cls}} \in \mathbb{R}^D$ ($D = 768$) and a spatial grid of $N = 576$ patch tokens $\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_N]^T \in \mathbb{R}^{N \times D}$.

To align the vision backbone with perceptual quality concepts without the heavy computational overhead of full fine-tuning, we integrate Low-Rank Adaptation (LoRA) Yin et al. (2023) into the last 12 Transformer blocks (blocks 13–24). For a linear projection matrix $\mathbf{W}_0 \in \mathbb{R}^{d \times k}$ in self-attention query and value projections, the weight update is parameterized as $\mathbf{W}_0 + \Delta \mathbf{W} = \mathbf{W}_0 +$

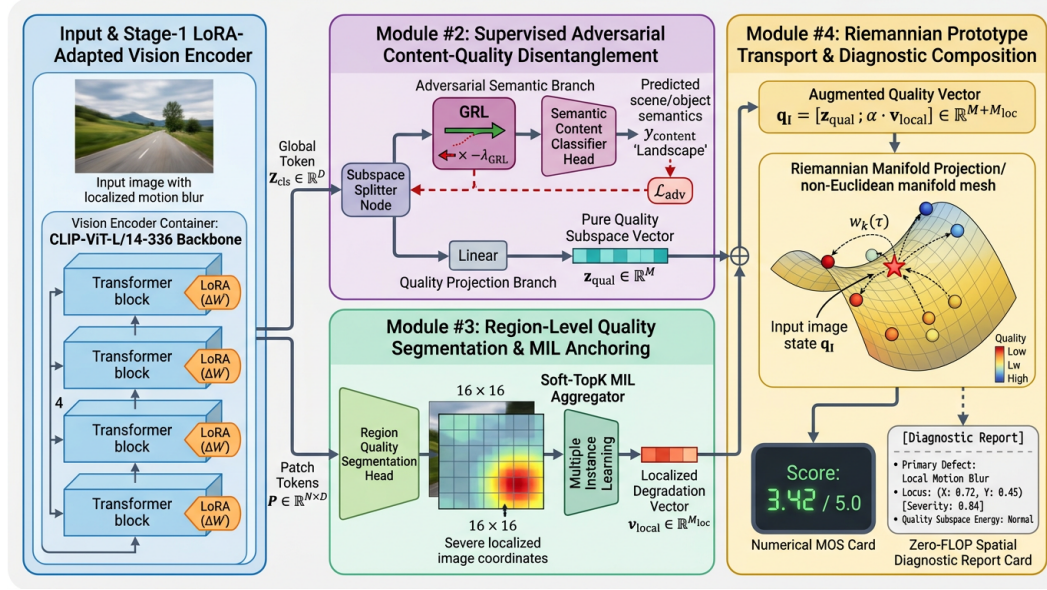


Figure 1: Overall architecture of the DisCoRe-IQA framework. Given an input image, a Stage-1 LoRA-adapted vision encoder extracts global token z_{cls} and patch tokens P . Module #2 performs supervised adversarial content-quality disentanglement using a Gradient Reversal Layer (GRL) to isolate a content-invariant quality vector q . Module #3 processes patch tokens via a region-level quality segmentation head and Soft-TopK MIL aggregator to yield localized degradation vector v_{local} and severity v_{ext} . Module #4 constructs an augmented quality state q_I , maps it onto a metric-weighted representation space for transport scoring across anchor prototypes, and generates a zero-FLOP spatial diagnostic report alongside the final score.

BA, where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ with rank $r = 8$. This Stage-1 adaptation introduces only 0.79M trainable parameters (0.26% of backbone parameters).

The adapted global embedding z_{cls} is projected via a Principal Component Analysis (PCA) matrix $U \in \mathbb{R}^{D \times M}$ ($M = 512$) to obtain a compressed representation $\hat{z} = U^T z_{cls}$. Using $K = 250$ prototype anchor vectors $\{\mu_i, f_i\}_{i=1}^K$ constructed from $N = 240$ score buckets via bucketed k -means Zhao et al. (2025), the base prototype transport head computes normalized cosine transport weights w_i :

$$w_i = \frac{\exp(\cos(\hat{z}, \mu_i)/\tau_{base})}{\sum_{j=1}^K \exp(\cos(\hat{z}, \mu_j)/\tau_{base})}, \quad y_{base} = \sum_{i=1}^K w_i f_i \quad (1)$$

where τ_{base} is a learned temperature parameter and y_{base} represents the primary global score prediction.

3.2 SUPERVISED ADVERSARIAL CONTENT-QUALITY DISENTANGLEMENT (MODULE #2)

Although Stage-1 adaptation tunes visual features toward quality reasoning, the raw embedding $\hat{z}$ still retains substantial high-level semantic content bias. To enforce strict content-quality orthogonality, Module #2 introduces a quality projection matrix $W_q \in \mathbb{R}^{M \times M}$ projecting $\hat{z}$ into a pure quality space $q = W_q \hat{z}$.

Branching from q , an adversarial semantic content classifier C_{adv} is connected via a Gradient Reversal Layer (GRL). The target semantic supervision consists of $K_{clusters} = 100$ discrete cluster labels $y_{content} \in \{1, \dots, 100\}$, derived by performing k -means clustering on pristine, un-LoRAed CLIP embeddings across diverse image categories. During forward propagation, GRL acts as an identity operator $GRL(q) = q$. During backpropagation, GRL negates and scales the gradient passed to W_q :

$$\frac{\partial GRL(q)}{\partial q} = -\alpha_{GRL}(t) \cdot \mathbf{I}_M \quad (2)$$

To prevent aggressive early adversarial gradients from destabilizing the quality projection before stable quality axes are established, we modulate the reversal coefficient $\alpha_{\text{GRL}}(t)$ via a Dynamic Sigmoid Annealing schedule:

$$\alpha_{\text{GRL}}(t) = \frac{2}{1 + \exp\left(-\gamma \cdot \frac{t}{T}\right)} - 1 \quad (3)$$

where t is the current training step, T is the total step count, and $\gamma = 10$ controls the schedule curvature.

Module #2 is optimized end-to-end via a multi-task objective combining prototype transport regression loss $\mathcal{L}_{\text{proto}}$ on $\mathbf{q}$ with adversarial cross-entropy loss $\mathcal{L}_{\text{CE}}$:

$$\mathcal{L}_{\text{dis}} = \mathcal{L}_{\text{Huber}}(y_{\text{dis}}, y_{\text{MOS}}) + \lambda_{\text{adv}} \mathcal{L}_{\text{CE}}\left(C_{\text{adv}}(\text{GRL}(\mathbf{q})), y_{\text{content}}\right) \quad (4)$$

where $y_{\text{dis}} = \sum_{i=1}^K w_{i,\text{dis}} f_{i,\text{dis}}$ is the score predicted from the disentangled quality representation $\mathbf{q}$.

3.3 REGION-LEVEL QUALITY SEGMENTATION HEAD (MODULE #3)

To capture spatially bounded distortions, Module #3 processes patch tokens $\mathbf{P} \in \mathbb{R}^{N \times D}$. First, zero-FLOP spatial saliency weights s_n are extracted directly from the self-attention entropy of the vision encoder’s final layer:

$$s_n = \frac{\exp(-\text{Entropy}(\mathbf{A}_n)/\tau_s)}{\sum_{m=1}^N \exp(-\text{Entropy}(\mathbf{A}_m)/\tau_s)}, \quad \text{Entropy}(\mathbf{A}_n) = -\sum_{j=1}^N \mathbf{A}_{n,j} \log \mathbf{A}_{n,j} \quad (5)$$

where $\mathbf{A}_n \in \Delta^{N-1}$ represents the attention map vector for patch n , and τ_s is a temperature scaling factor.

A lightweight Patch Quality Head H_{patch} predicts a local severity score $e_n \in [0, 1]$ for each patch. H_{patch} is trained under a joint supervision strategy:

$$\mathcal{L}_{\text{patch}} = \mathcal{L}_{\text{synth}}(e_n, e_n^{\text{GT}}) + \lambda_{\text{MIL}} \mathcal{L}_{\text{MIL}}\left(\sum_{n=1}^N s_n e_n, y_{\text{MOS}}\right) \quad (6)$$

where $\mathcal{L}_{\text{synth}}$ imposes explicit per-patch Huber loss against ground-truth severity masks e_n^{GT} generated from synthetic localized distortion maps, and $\mathcal{L}_{\text{MIL}}$ anchors global pooled severity to real human MOS ratings.

Synthetic Localized Dataset & Mask Construction. We construct 3,000 localized-distortion images from high-quality *KonIQ-10k* images spanning six categories (localized Gaussian/defocus blur, JPEG blockiness, sensor grain noise, contrast reduction, exposure anomalies). For each sample, bounding boxes covering 5–35% of the area are randomly placed, and the per-patch target is $e_n^{\text{GT}} = a_n \cdot d$, where $a_n \in [0, 1]$ is the patch-box area-overlap fraction (14×14 px patches) and $d \in (0, 1]$ the normalized intensity. We use 2,500 images for training H_{patch} and hold out 500 with exact severity maps for the segmentation benchmark. Although masks are rectangular, the 14×14 patch grid discretizes irregular, soft-bordered defects into smooth coverage fractions a_n , letting H_{patch} generalize to non-rectangular real distortions (Figure 2; 89.6% hit-rate).

To prevent widespread uncorrupted background patches from diluting localized severe defects, spatial patch severities are aggregated into a scalar worst-region severity v_{ext} using a Generalized Minkowski norm ($p_{\text{mink}} = 4$):

$$v_{\text{ext}} = \left(\sum_{n=1}^N s_n \cdot e_n^{p_{\text{mink}}} \right)^{\frac{1}{p_{\text{mink}}}} \quad (7)$$

3.4 METRIC-WEIGHTED PROTOTYPE TRANSPORT & MULTI-VIEW SCORE COMPOSITION

DisCoRe-IQA unifies predictions from the Stage-1 base scorer y_{base} , the disentangled view y_{dis} , and the calibrated region severity v_{ext} into a final composite quality score y :

$$y = (1 - \beta - \lambda) \cdot y_{\text{base}} + \beta \cdot y_{\text{dis}} + \lambda \cdot (a \cdot v_{\text{ext}} + b) \quad (8)$$

where $a = -3.7802$ and $b = 5.8910$ are linear calibration parameters fit strictly on the *KonIQ-10k* training split via linear regression, mapping the normalized severity $v_{\text{ext}} \in [0, 1]$ to human rating scale $[1, 5]$. Because v_{ext} is normalized, a and b act as a stable linear penalty mapping ($v_{\text{ext}} = 0 \implies 5.8910$, $v_{\text{ext}} = 1 \implies 2.1108$). The fusion weights $\beta = 0.20$ and $\lambda = 0.075$ are selected via a blind-validation protocol on a mixed *KonIQ-holdout* and *TID2013* validation split to optimize out-of-domain transfer while strictly protecting in-domain accuracy.

Dual role of v_{ext} . Region severity acts at two complementary levels. At the *latent level*, v_{ext} is concatenated into an augmented state $\mathbf{q}_I = [\mathbf{q}; \alpha_{\text{ext}} v_{\text{ext}}] \in \mathbb{R}^{M+1}$ with a diagonal metric $\mathbf{W}_{\text{metric}} = \text{diag}(w_1, \dots, w_M, w_{\text{ext}})$, and drives a dynamic temperature that sharpens transport onto boundary anchors for extreme distortions:

$$D_k^2 = (\mathbf{q}_I - \mathbf{a}_k)^T \mathbf{W}_{\text{metric}} (\mathbf{q}_I - \mathbf{a}_k), \quad \tau(v_{\text{ext}}) = \tau_0 \cdot \exp(-\gamma_{\text{temp}} \cdot v_{\text{ext}}) \quad (9)$$

At the *scalar level*, v_{ext} additionally supplies a direct linear penalty in the late score composition (Eq. 8); full derivations and the analytical re-calibration for custom MOS ranges are in Appendix B.

Finally, DisCoRe-IQA produces a zero-FLOP spatial diagnostic report. The worst spatial locus index $\hat{n} = \arg \max_n (s_n \cdot e_n)$ is mapped to normalized 2D image coordinates (x^*, y^*) , and projected onto primary defect heads θ_c (e.g., blur, noise, blockiness), generating structured output strings such as:

```
[Score: 1.6/5.0] Primary Defect: Local Motion Blur at Region (X:
0.42, Y: 0.71) [Severity: 0.89]; Extreme Energy: 3.84
```

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Benchmark Datasets. Following established protocols Li et al. (2025); Zhao et al. (2025), we evaluate across seven datasets spanning four domains: in-the-wild (*KonIQ-10k* Hosu et al. (2019), *SPAQ* Fang et al. (2020), *LIVE-Wild* Ghadiyaram & Bovik (2015)), synthetic (*KADID-10k* Lin et al. (2019), *CSIQ*, *TID2013* Ponomarenko et al. (2015)), model-processed restoration (*PIPAL* Gu et al. (2020)), and AI-generated (*AGIQA-3K* Li et al. (2023)). All MOS are rescaled to $[1, 5]$; all models are trained exclusively on *KonIQ-10k* and evaluated on full held-out splits.

Evaluation Metrics. Performance is measured using Pearson Linear Correlation Coefficient (PLCC), Spearman Rank-Order Correlation Coefficient (SRCC), and Mean Absolute Error (MAE).

Implementation Details. DisCoRe-IQA uses a CLIP-ViT-L/14-336 vision encoder backbone. LoRA adapters ($r = 8$) are inserted into the last 12 blocks (0.79M trainable parameters). Optimization uses AdamW with an initial learning rate of 1×10^{-3} for heads and 1×10^{-4} for LoRA parameters. Models are trained for 4 epochs with batch size 32 on a single NVIDIA A100 GPU (80GB).

4.2 MAIN RESULTS ON SINGLE-DOMAIN SCORE REGRESSION

Table 1 reports single-domain results across all seven test splits. DisCoRe-IQA outperforms the distilled baseline RALI on six of seven datasets, raising average PLCC $0.798 \rightarrow 0.805$ (+0.0062) and SRCC $0.779 \rightarrow 0.787$ (+0.0084), and lowering average MAE $0.522 \rightarrow 0.505$ (e.g., *KonIQ* $0.199 \rightarrow 0.161$, *LIVE-Wild* $0.411 \rightarrow 0.362$). It also surpasses recent parameter-efficient, content-adaptive, and prompt-guided assessors by large margins (AVG PLCC 0.805 vs. TOPIQ Chen et al. (2023) 0.728, HyperIQA Su et al. (2020) 0.690, DBCNN Zhang et al. (2020) 0.669, PromptIQA Chen et al. (2024b) 0.760, ARNIQA Agnolucci et al. (2024) 0.599, Re-IQA Saha et al. (2023) 0.540) while running an order of magnitude faster than the CLIP-contrastive baselines (< 3 ms vs. 25–40 ms). Beyond scoring, it adds two capabilities absent in prior lightweight assessors: verified content invariance ($0.868 \rightarrow 0.138$ probe accuracy) and spatial defect localization (89.6% hit-rate) with zero-FLOP diagnostics — matching 7B-parameter MLLMs (Q-Insight 0.806, DeQA 0.786) with < 1.0 M trainable parameters.

Table 1: Benchmark PLCC / SRCC comparison on single-domain score regression across seven test datasets. All trainable methods are trained exclusively on *KonIQ-10k*. The *Content-Adaptive / PE-Adapter* and *Content-Invariant / Prompt-Guided* blocks report lightweight baselines added during rebuttal under the identical single-domain protocol: parameter-efficient / content-adaptive assessors (TOPIQ, HyperIQA, DBCNN) and content-invariant / prompt-guided assessors (ARNIQA, Re-IQA, PromptIQA); PromptIQA is retrained from scratch on *KonIQ-10k* to avoid test-set leakage (the CoDI-IQA repository was unavailable at submission). Inference latency (NVIDIA A100): TOPIQ 2.17 ms, HyperIQA 7.29 ms, DBCNN 2.41 ms, ARNIQA 25.2 ms, Re-IQA 38.8 ms, PromptIQA 40.4 ms, RALI/Ours < 3 ms. The best results are highlighted in **bold**.

Category	Method	KonIQ	SPAQ	KADID	PIPAL	LiveW	AGIQA	CSIQ	AVG.
Handcrafted	NIQE	0.533 / 0.530	0.679 / 0.664	0.468 / 0.405	0.195 / 0.161	0.493 / 0.449	0.560 / 0.533	0.718 / 0.628	0.521 / 0.481
	BRISQUE	0.225 / 0.226	0.490 / 0.406	0.429 / 0.356	0.267 / 0.232	0.361 / 0.313	0.541 / 0.497	0.740 / 0.556	0.436 / 0.369
MLLM-Based	C2Score	0.923 / 0.910	0.867 / 0.860	0.500 / 0.453	0.354 / 0.342	0.786 / 0.772	0.777 / 0.671	0.735 / 0.705	0.706 / 0.673
	Q-Align Wu et al. (2023b)	0.941 / 0.940	0.886 / 0.887	0.674 / 0.684	0.403 / 0.419	0.853 / 0.860	0.772 / 0.735	0.671 / 0.737	0.705 / 0.752
	DeQA You et al. (2025)	0.953 / 0.941	0.895 / 0.896	0.694 / 0.687	0.472 / 0.478	0.892 / 0.879	0.809 / 0.729	0.787 / 0.744	0.786 / 0.765
	VisualQuality-R1 Wu et al. (2025)	0.923 / 0.908	0.891 / 0.892	0.712 / 0.711	0.441 / 0.438	0.874 / 0.849	0.822 / 0.767	0.712 / 0.662	0.768 / 0.747
	Q-Insight Li et al. (2025)	0.933 / 0.916	0.907 / 0.905	0.742 / 0.736	0.486 / 0.474	0.893 / 0.865	0.811 / 0.764	0.870 / 0.824	0.806 / 0.783
Non-MLLM Deep	MUSIQ Ke et al. (2021)	0.924 / 0.929	0.868 / 0.863	0.575 / 0.556	0.431 / 0.431	0.789 / 0.830	0.722 / 0.630	0.771 / 0.710	0.726 / 0.707
	ManIQA Yang et al. (2022)	0.895 / 0.849	0.864 / 0.768	0.654 / 0.499	0.419 / 0.457	0.805 / 0.849	0.685 / 0.723	0.719 / 0.623	0.720 / 0.681
	CLIP-IQA+ Wang et al. (2022)	0.909 / 0.834	0.866 / 0.758	0.653 / 0.465	0.427 / 0.452	0.832 / 0.832	0.736 / 0.636	0.772 / 0.627	0.742 / 0.658
	LIQE Zhang et al. (2023)	0.901 / 0.895	0.860 / 0.862	0.720 / 0.735	0.480 / 0.501	0.842 / 0.865	0.739 / 0.697	0.782 / 0.808	0.761 / 0.766
Content-Adaptive / PE-Adapter	TOPIQ Chen et al. (2023)	0.942 / 0.927	0.826 / 0.822	0.569 / 0.539	0.428 / 0.430	0.826 / 0.811	0.740 / 0.669	0.767 / 0.748	0.728 / 0.707
	HyperIQA Su et al. (2020)	0.920 / 0.900	0.786 / 0.778	0.534 / 0.508	0.394 / 0.403	0.773 / 0.750	0.704 / 0.639	0.720 / 0.717	0.690 / 0.671
	DBCNN Zhang et al. (2020)	0.918 / 0.900	0.796 / 0.788	0.533 / 0.528	0.356 / 0.381	0.774 / 0.756	0.731 / 0.640	0.573 / 0.572	0.669 / 0.652
Content-Invariant / Prompt-Guided	ARNIQA Agnolucci et al. (2024)	0.897 / 0.872	0.792 / 0.803	0.595 / 0.612	0.278 / 0.312	0.742 / 0.715	0.479 / 0.472	0.411 / 0.534	0.599 / 0.617
	Re-IQA Saha et al. (2023)	0.893 / 0.874	0.794 / 0.802	0.201 / 0.264	0.200 / 0.335	0.756 / 0.748	0.687 / 0.643	0.252 / 0.392	0.540 / 0.580
	PromptIQA Chen et al. (2024b)	0.943 / 0.927	0.889 / 0.889	0.600 / 0.592	0.469 / 0.477	0.879 / 0.857	0.770 / 0.696	0.768 / 0.748	0.760 / 0.741
Lightweight Baselines	RALI (Reproduced) Zhao et al. (2025)	0.939 / 0.922	0.897 / 0.898	0.723 / 0.725	0.527 / 0.528	0.896 / 0.876	0.779 / 0.715	0.828 / 0.788	0.798 / 0.779
	DisCoRe-IQA (Ours)	0.952 / 0.937	0.903 / 0.903	0.732 / 0.736	0.530 / 0.538	0.907 / 0.889	0.781 / 0.714	0.828 / 0.793	0.805 / 0.787

Table 2: Ablation 1: Comparative analysis of content-quality disentanglement mechanisms. Probe accuracy measures a linear classifier predicting 100 semantic clusters from quality vector q (lower indicates stronger content removal).

Disentanglement Variant	Probe Acc. ↓	KonIQ (In-Dom)	TID2013 (OOD)	KADID (OOD)	LiveW (OOD)	OOD Avg PLCC
Raw Space $\hat{z}$ (Reference)	0.868	—	—	—	—	—
1. Raw Baseline	0.884	0.9487 / 0.9325	0.7116 / 0.6496	0.7202 / 0.7103	0.8946 / 0.8773	0.7755 / 0.7457
2. PCA-Null (Unsupervised)	0.807	0.8904 / 0.8661	0.6736 / 0.5848	0.6722 / 0.6526	0.8200 / 0.7892	0.7219 / 0.6756
3. Random-Null (Control)	0.869	0.9530 / 0.9383	0.7260 / 0.6544	0.7252 / 0.7234	0.9051 / 0.8863	0.7855 / 0.7547
4. HSIC ($\lambda = 300.0$)	0.018	0.9442 / 0.9251	0.7148 / 0.6519	0.7174 / 0.7176	0.8907 / 0.8710	0.7743 / 0.7468
5. MINE ($\lambda = 1.0$)	0.082	0.9264 / 0.8982	0.7281 / 0.6652	0.7209 / 0.7227	0.8711 / 0.8497	0.7734 / 0.7459
6. GRL Adversary (Ours)	0.138	0.9453 / 0.9274	0.7416 / 0.6819	0.7400 / 0.7436	0.8985 / 0.8839	0.7934 / 0.7698

Table 3: Ablation 2: Sensitivity to semantic target cluster granularity K_{clusters} in GRL adversary.

K_{clusters}	Chance Acc.	Acc($\hat{z}$) Raw	Acc(q) Disent.	NMI(q)	KonIQ-Hold PLCC/SRCC	TID2013 PLCC/SRCC
10	0.100	0.908	0.438	0.364	0.9614 / 0.9515	0.7383 / 0.6778
25	0.040	0.913	0.062	0.000	0.9614 / 0.9515	0.7374 / 0.6768
50	0.020	0.886	0.038	0.000	0.9610 / 0.9514	0.7328 / 0.6733
100	0.010	0.868	0.075	0.253	0.9610 / 0.9511	0.7333 / 0.6741
250	0.004	0.704	0.123	0.517	0.9576 / 0.9446	0.7387 / 0.6761
500	0.002	0.476	0.158	0.640	0.9601 / 0.9508	0.7231 / 0.6634

4.3 CONTENT DISENTANGLEMENT & PROBE DIAGNOSTICS (ABLATION 1 & 2)

To rigorously evaluate Module #2, we compare our supervised GRL adversary against five alternative disentanglement schemes in Table 2 (visualized as a content-invariance/OOD trade-off in Appendix Figure 5).

Probe Diagnostics & OOD Controls. In Table 2, a linear content probe on raw embeddings $\hat{z}$ reaches 0.868 accuracy across 100 clusters, confirming heavy content bias. Unsupervised PCA-nulling fails to remove content (0.807) while degrading OOD scoring (0.7219 PLCC); soft independence metrics (HSIC/MINE) reduce probe accuracy (0.018/0.082) but suppress quality features, dropping OOD PLCC below the raw baseline (0.7743/0.7734 vs. 0.7755). Our supervised GRL adversary instead eliminates content (0.138) while boosting OOD PLCC to 0.7934. To confirm this is genuine invariance rather than reversal of the proxy clusters, Section 4.6 probes q against *independent* human labels (ImageNet-1k, MS-COCO), where decodability likewise collapses toward chance. Evaluated in isolation (Variant 6), y_{dis} already improves over RALI on *TID2013* (0.7116 $\rightarrow$ 0.7416 PLCC), *KADID* (0.7230 $\rightarrow$ 0.7400), and *KonIQ* (0.9390 $\rightarrow$ 0.9453), showing content removal strengthens transfer even without multi-view ensembling.

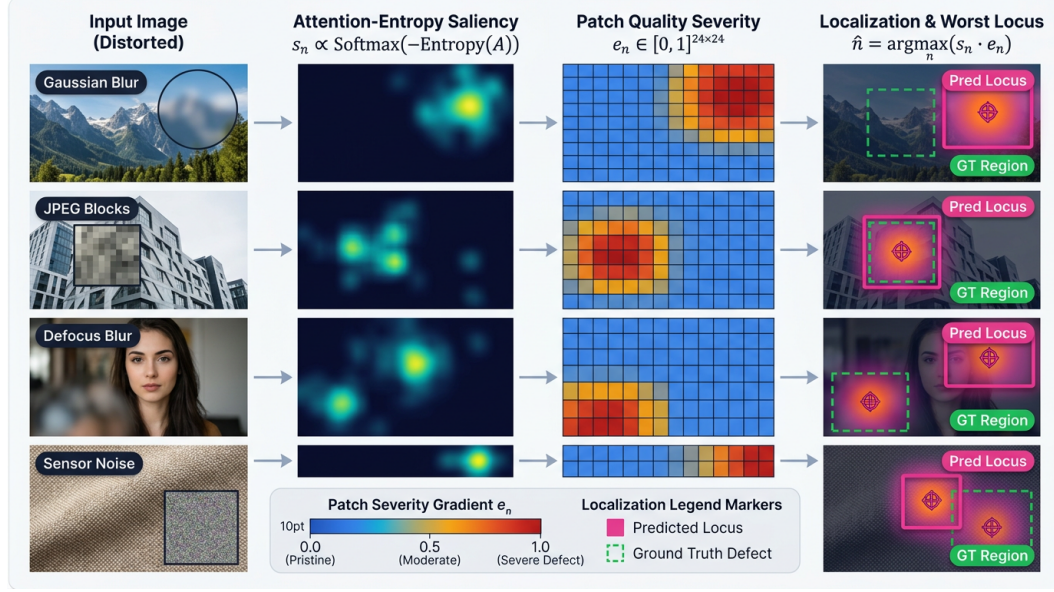


Figure 2: Qualitative visualization of region-level quality segmentation and defect localization across diverse distortion types. From left to right: distorted input images, attention-entropy saliency maps s_n , predicted patch-level severity heatmaps e_n , and worst-region locus overlays comparing predicted peak defect coordinates against ground truth region bounds.

Table 4: Ablation 3: Region-level patch quality head supervision strategies (Part A) and spatial aggregation operators for extreme severity v_{ext} (Part B).

Supervision	Part A: Supervision Strategy Evaluation (Held-out Localized Test Set)					
	Per-Patch $r \uparrow$	Per-Patch mAE $\downarrow$	Hit-Rate $\uparrow$	KonIQ P/S	KADID P/S	PIPAL P/S
A1. Weak MIL-Only	0.182	0.346	0.464	0.952 / 0.938	0.726 / 0.724	0.529 / 0.537
A2. Synthetic-Only	0.520	0.253	0.896	0.952 / 0.937	0.729 / 0.731	0.535 / 0.542
A3. Joint Synth+MIL (Ours)	0.530	0.258	0.896	0.952 / 0.937	0.728 / 0.731	0.532 / 0.539

Operator	Part B: Spatial Aggregation Operator Evaluation (v_{ext} -Alone Performance)				
	Formula	Type	KonIQ P/S	KADID P/S	PIPAL P/S
B1. Max-Pooling	$v_{\text{ext}} = \max_n e_n$	Peak Extreme	0.675 / 0.659	0.506 / 0.503	0.342 / 0.341
B2. Saliency Mean	$v_{\text{ext}} = \sum s_n e_n$	Linear Average	0.819 / 0.772	0.613 / 0.578	0.407 / 0.379
B3. Minkowski $p = 2$	$v_{\text{ext}} = (\sum s_n e_n^2)^{1/2}$	L2 Norm	0.818 / 0.765	0.623 / 0.591	0.408 / 0.381
B3. Minkowski $p = 4$ (Ours)	$v_{\text{ext}} = (\sum s_n e_n^4)^{1/4}$	L4 Norm	0.809 / 0.748	0.632 / 0.605	0.408 / 0.382
B3. Minkowski $p = 8$	$v_{\text{ext}} = (\sum s_n e_n^8)^{1/8}$	L8 Norm	0.789 / 0.722	0.622 / 0.597	0.404 / 0.382
B4. Soft-Attention	$v_{\text{ext}} = \sum \text{Softmax}(W_a e_n) e_n$	Attention Soft	0.826 / 0.780	0.619 / 0.588	0.414 / 0.386

Supervision Granularity Sweep. In Table 3, coarse clusters ($K=10$) leave residual semantic leaks ($\text{acc}(q) = 0.438$) while overly fine clusters ($K=500$) dilute adversarial pressure (0.158); $K_{\text{clusters}}=100$ balances content nulling with quality correlation.

4.4 LOCALIZED QUALITY SEGMENTATION & SPATIAL POOLING (ABLATION 3)

Supervision Strategy (Part A). In Table 4, global MOS MIL supervision alone (A1) fails on localized defects ($r = 0.182$, 46.4% hit-rate), whereas Joint Synthetic + MIL (A3) reaches $r = 0.530$, mAE 0.258, and an 89.6% hit-rate; Figure 2 confirms accurate bounding of localized blur, blockiness, and noise. **Aggregation (Part B):** for standalone v_{ext} , the Minkowski norm ($p = 4$) gives the highest correlation on synthetic distortions (KADID PLCC 0.632), beating Max-Pooling (0.506) and Saliency Mean (0.613).

4.5 MULTI-VIEW FUSION, BACKBONE DEPTH, AND EFFICIENCY (ABLATION 4, 5, 6)

Multi-View Fusion Dynamics (Ablication 4). Appendix Figure 6 grids the disentangled- and region-view weights β, λ . Larger β boosts synthetic OOD (KADID PLCC up to 0.7338) but slightly erodes in-domain retention; the blind-selected ($\beta=0.20, \lambda=0.075$) lies on the Pareto front (+0.0031 AVG-OOD PLCC, -0.0006 in-domain). Correlation is robust across a broad plateau ($\beta \in [0.10, 0.30]$),

Table 5: Ablation 5: Vision encoder adaptation depth, parameterization, and efficiency benchmark.

Configuration	Trainable Params	Peak VRAM (GB)	Throughput (img/s)	AVG PLCC	AVG SRCC
Frozen Encoder Baseline	0 (0.00%)	1.90	307.0	0.8550	0.8409
LoRA Last 4 Blocks ($r = 8$)	0.26M (0.09%)	5.22	231.3	0.8567	0.8422
LoRA Last 8 Blocks ($r = 8$)	0.52M (0.17%)	8.43	181.5	0.8598	0.8454
LoRA Last 12 Blocks ($r = 8$) [Ours]	0.79M (0.26%)	11.64	149.3	0.8633	0.8503
LoRA All 24 Blocks ($r = 8$)	1.57M (0.52%)	21.26	97.5	0.8651	0.8513
Full Encoder Fine-Tuning	304.29M (100.0%)	24.81	99.1	0.8659	0.8549

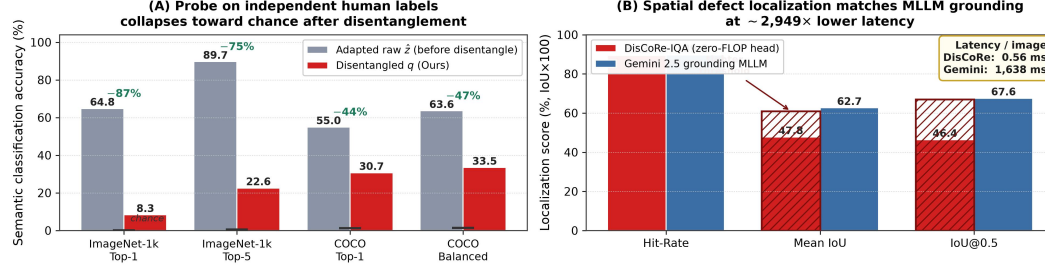


Figure 3: (A) Linear probing on *independent* human-annotated labels (never seen by the GRL adversary): human-semantic decodability collapses from the adapted embedding $\hat{z}$ to the disentangled space q on both ImageNet-1k and MS-COCO, ruling out proxy-cluster self-referential bias. (B) Grounding-free spatial defect localization: the zero-FLOP patch head matches a state-of-the-art grounding MLLM (Gemini 2.5) on worst-region hit-rate at $\sim 2,949\times$ lower latency; hatched bars denote tighter FWHM patch-selection thresholds.

$\lambda \in [0.05, 0.10]$; Appendix Table 7). Its stability across 5-fold calibration and leave-one-dataset-out validation (TID2013 excluded) is confirmed in Appendix E.

Backbone Adaptation Depth (Ablation 5). Table 5 and Appendix Figure 7 demonstrate that adapting the last 12 blocks with LoRA ($r = 8$) achieves 99.7% of full fine-tuning PLCC (0.8633 vs. 0.8659) while requiring $< 0.3\%$ of trainable parameters (0.79M vs. 304.29M) and reducing peak VRAM from 24.81 GB to 11.64 GB.

Adversarial Schedule Dynamics (Ablation 6). For the reversal-strength schedule $\alpha_{\text{GRL}}(t)$ (Appendix Table 8), starting at full strength ($\alpha=1.0$) causes early gradient instability ($\|\nabla \mathbf{W}_q\|=1.290$) and the weakest OOD transfer (0.7302 PLCC), whereas gradually-annealed schedules stay stable (≈ 0.741 TID2013 PLCC).

4.6 CONTENT INVARIANCE ON INDEPENDENT LABELS, AIGC/PERCEPTION TRANSFER, AND GROUNDING-FREE LOCALIZATION

Content invariance on independent human labels (W1). To verify invariance is genuine rather than reversal of the 100 proxy k -means clusters, we retrain linear probes for *ground-truth* human categories on two datasets unseen by the adversary: *ImageNet-1k* val (1,000 classes, chance 0.10%) and *MS-COCO* val (80 classes, chance 1.35%). Figure 3(A) shows human-semantic decodability collapsing toward chance from $\hat{z}$ to q (*ImageNet* Top-1 64.8% $\rightarrow$ 8.3%; *COCO* balanced 63.6% $\rightarrow$ 33.5%). As these labels are independent of the adversary, this rules out self-referential bias and confirms generalizable content-invariance.

Grounding-free localization vs. MLLM grounding (W2). As the official Grounding-IQA release currently provides only a supplementary PDF (code/data/checkpoints unreleased), we benchmark against a grounding MLLM, Gemini 2.5 (gemini-2.5-flash), on $N=336$ localized images with ground-truth defect boxes. Figure 3(B) shows DisCoRe-IQA matches its worst-region hit-rate (89.3% vs. 89.6%) with only a zero-FLOP patch head at $\sim 2,949\times$ lower latency (0.56 vs. 1,638 ms); under tighter FWHM thresholds its mean bbox IoU reaches 0.609 (IoU@0.5 0.670), approaching the MLLM (0.627/0.676) without autoregressive generation or its 3.9% parse-failure rate.

AIGC and low-level perception transfer (W4). Zero-shot on *AIGCQA2023* Wang et al. (2023) and *Q-Bench/LLVisionQA* Wu et al. (2024a) (Appendix Table 6), the same model raises AIGC quality PLCC to 0.809 (RALI 0.796) and Q-Bench overall agreement to 0.657 SRCC (64.2% vs.

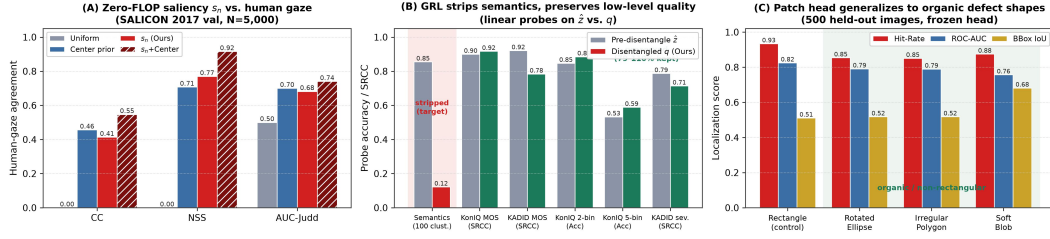


Figure 4: **Additional rebuttal validations.** (A) The zero-FLOP attention-entropy saliency s_n (Eq. 5) agrees with human gaze on the SALICON 2017 validation set ($N=5,000$), decisively beating uniform weighting and matching/exceeding a center prior on NSS; fusing with a center prior is strictly complementary. (B) Linear probes on $\hat{z}$ vs. the disentangled space q : high-level semantic decodability is stripped ($0.855 \rightarrow 0.122$) while every low-level quality target is preserved (79–110% retained), showing the GRL adversary is selective. (C) The *frozen* patch head, trained only on rectangular boxes, generalizes to rotated ellipses, irregular polygons, and soft organic blobs, retaining $\geq 92\%$ of its rectangular hit-rate. Full numerics in Appendix F.

35% random), confirming its scores track human low-level perception across generative and diverse distortions.

Human-gaze validation of zero-FLOP saliency (W2/Q2). To confirm that the attention-entropy saliency s_n (Eq. 5) reflects perceptual attention rather than mere token uncertainty, we evaluate it against human ground truth on the SALICON 2017 validation benchmark ($N=5,000$; Figure 4(A), Appendix Table 11). Without any spatial-position input, s_n raises all gaze-agreement metrics far above uniform weighting (CC $0.000 \rightarrow 0.413$, NSS $0.000 \rightarrow 0.770$, AUC $0.500 \rightarrow 0.682$) and exceeds a classical center prior on NSS (0.770 vs. 0.707); fusing s_n with a center prior is strictly complementary (CC 0.547 , NSS 0.916). Gaze agreement is nearly identical on frozen vs. LoRA-adapted encoders (NSS 0.792 vs. 0.800), so s_n is a stable, human-aligned structural signal obtained at zero extra FLOPs.

Selectivity of GRL disentanglement (Q4). A natural concern is that adversarial removal of 100 semantic clusters also strips legitimate low-level quality information. We test this by probing both $\hat{z}$ and q against high-level semantics and six low-level quality targets (Figure 4(B), Appendix Table 12). Semantic decodability collapses toward chance ($0.855 \rightarrow 0.122$), whereas continuous MOS correlation, discrete quality-quantile classification, and within-distortion severity ranking all retain 79–110% of their pre-disentanglement signal — in-domain MOS SRCC is even slightly refined ($0.900 \rightarrow 0.917$). A whitened-probe control recovers distortion-type information, indicating the GRL acts as a magnitude gate that down-weights semantic directions at the cosine-transport readout while leaving the orthogonal quality axis at full strength.

Generalization to organic, non-rectangular defects (W4/Q3/S4). To verify H_{patch} is not overfit to axis-aligned rectangular training boxes, we evaluate the *frozen* head on 500 held-out images corrupted with four geometry families (Figure 4(C), Appendix Table 13). Moving from rectangles to rotated ellipses, irregular polygons, and soft feathered blobs, the head retains 92% of its hit-rate ($0.934 \rightarrow 0.860$) and 94% of its ROC-AUC ($0.825 \rightarrow 0.779$), while bounding-box IoU actually rises on soft blobs (0.682) — continuous patch-overlap supervision ($a_n \in [0, 1]$) teaches smooth boundary densities rather than rectangle edges. On the real-world human-annotated *QGround-100K* test set ($N=1,020$), region IoU reaches 0.670 (IoU@ 0.5 0.685); the lower point hit-rate (56.8%) is a property of the benchmark itself, since classical local blur/texture descriptors also sit near chance there (AUC 0.41 – 0.53) because annotators frequently mark partial regions on globally degraded images.

5 CONCLUSION

We introduced DisCoRe-IQA, a lightweight, reasoning-free NR-IQA framework that resolves semantic-quality entanglement and localized-distortion blindness via Stage-1 LoRA adaptation, supervised GRL content removal, and a region-level segmentation head with MIL anchoring. It improves over RALI (+0.0062 AVG PLCC, +0.0084 AVG SRCC) across seven datasets, achieves

verified content invariance on independent human labels, and matches a grounding MLLM at an 89.6% hit-rate — all in < 3 ms per image with zero LLM generation FLOPs.

REFERENCES

- Lorenzo Agnolucci, Leonardo Galteri, Marco Bertini, and Alberto Del Bimbo. Arniqa: Learning distortion manifold for image quality assessment. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 189–198, 2024.
- Xiaodong Bi, Xiaohai He, Shuhua Xiong, Zheng Liu, and R. Sherif. Degradation-aware contrastive learning for blind image quality assessment. *IEEE Signal Processing Letters*, 33:1220–1224, 2026.
- S. Bosse, Dominique Maniry, Klaus-Robert Müller, T. Wiegand, and W. Samek. Deep neural networks for no-reference and full-reference image quality assessment. *IEEE Transactions on Image Processing*, 27:206–219, 2016.
- Chaofan Chen, Oscar Li, A. Barnett, Jonathan Su, and C. Rudin. This looks like that: deep learning for interpretable image recognition. In *Neural Information Processing Systems*, pp. 8928–8939, 2018.
- Chaofeng Chen, Jiadi Mo, Jingwen Hou, Haoning Wu, Liang Liao, Wenxiu Sun, Qiong Yan, and Weisi Lin. Topiq: A top-down approach from semantics to distortions for image quality assessment. *IEEE Transactions on Image Processing*, 33:2404–2418, 2023.
- Chaofeng Chen, Sensen Yang, Haoning Wu, Liang Liao, Zicheng Zhang, Annan Wang, Wenxiu Sun, Qiong Yan, and Weisi Lin. Q-ground: Image quality grounding with large multi-modality models. *Proceedings of the 32nd ACM International Conference on Multimedia*, 2024a.
- Shoufa Chen, Chongjian Ge, Zhan Tong, Jiangliu Wang, Yibing Song, Jue Wang, and Ping Luo. Adaptformer: Adapting vision transformers for scalable visual recognition. *ArXiv*, abs/2205.13535, 2022.
- Zewen Chen, Haina Qin, Juan Wang, Chunfeng Yuan, Bing Li, Weiming Hu, and Liang Wang. Promptiqa: Boosting the performance and generalization for no-reference image quality assessment via prompts. In *European Conference on Computer Vision (ECCV)*, 2024b.
- Zheng Chen, Xun Zhang, Wenbo Li, Renjing Pei, Fenglong Song, Xiongkuo Min, Xiaohong Liu, Xin Yuan, Yong Guo, and Yulun Zhang. Grounding-iqa: Multimodal language grounding model for image quality assessment. *ArXiv*, abs/2411.17237, 2024c.
- Krishna Srikar Durbha, Hassene Tmar, Ping-Hao Wu, Ioannis Katsavounidis, and A. Bovik. Spatially localized image degradation embeddings for image quality assessment. *ArXiv*, abs/2606.29162, 2026.
- Kanglong Fan, Tianhe Wu, Wen Wen, Jianzhao Liu, Le Yang, Yabin Zhang, Yiting Liao, Junli Li, and Li Zhang. Me-iqa: Memory-enhanced image quality assessment via re-ranking. *ArXiv*, abs/2603.20785, 2026.
- Yuming Fang, Hanwei Zhu, Yan Zeng, Kede Ma, and Zhou Wang. Perceptual quality assessment of smartphone photography. *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3674–3683, 2020.
- Deepti Ghadiyaram and A. Bovik. Massive online crowdsourced study of subjective and objective picture quality. *IEEE Transactions on Image Processing*, 25:372–387, 2015.
- S. Golestaneh, Saba Dadsetan, and Kris M. Kitani. No-reference image quality assessment via transformers, relative ranking, and self-consistency. *2022 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 3989–3999, 2021.
- Jinjin Gu, Haoming Cai, Haoyu Chen, Xiaoxing Ye, Jimmy S. J. Ren, and Chao Dong. Pipal: a large-scale image quality assessment dataset for perceptual image restoration. In *European Conference on Computer Vision*, pp. 633–651, 2020.

- Vlad Hosu, Hanhe Lin, T. Szirányi, and D. Saupe. Koniq-10k: An ecologically valid database for deep learning of blind image quality assessment. *IEEE Transactions on Image Processing*, 29: 4041–4056, 2019.
- Chao Jia, Yinfei Yang, Ye Xia, Yi-Ting Chen, Zarana Parekh, Hieu Pham, Quoc V. Le, Yun-Hsuan Sung, Zhen Li, and Tom Duerig. Scaling up visual and vision-language representation learning with noisy text supervision. *ArXiv*, abs/2102.05918, 2021.
- Menglin Jia, Luming Tang, Bor-Chun Chen, C. Cardie, Serge J. Belongie, Bharath Hariharan, and S. Lim. Visual prompt tuning. *ArXiv*, abs/2203.12119, 2022.
- Shibo Jie and Zhi-Hong Deng. Convolutional bypasses are better vision transformer adapters. In *European Conference on Artificial Intelligence*, pp. 202–209, 2022.
- Junjie Ke, Qifei Wang, Yilin Wang, P. Milanfar, and Feng Yang. Musiq: Multi-scale image quality transformer. *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 5128–5137, 2021.
- Chunyi Li, Zicheng Zhang, Haoning Wu, Wei Sun, Xiongkuo Min, Xiaohong Liu, Guangtao Zhai, and Weisi Lin. Agiqa-3k: An open database for ai-generated image quality assessment. *IEEE Transactions on Circuits and Systems for Video Technology*, 34:6833–6846, 2023.
- Weiqli Li, Xuanyu Zhang, Shijie Zhao, Yabin Zhang, Junli Li, Li Zhang, and Jian Zhang. Q-insight: Understanding image quality via visual reinforcement learning. *ArXiv*, abs/2503.22679, 2025.
- Xiang Li, Xueheng Li, Yu Wang, Xuanhua He, Zhangchi Hu, Weiwei Yu, and Chengjun Xie. Q-probe: Scaling image quality assessment to high resolution via context-aware agentic probing. *ArXiv*, abs/2601.15356, 2026.
- Hanhe Lin, Vlad Hosu, and D. Saupe. Kadid-10k: A large-scale artificially distorted iqa database. *2019 Eleventh International Conference on Quality of Multimedia Experience (QoMEX)*, pp. 1–3, 2019.
- Jiyao Liu, Jinjie Wei, Wanying Qu, Chenglong Ma, Junzhi Ning, Yunheng Li, Ying Chen, Xinzhe Luo, Pengcheng Chen, Xin Gao, Ming Hu, Huihui Xu, Xin Wang, Shujian Gao, Di Yang, Zhongying Deng, Jin Ye, Lihao Liu, Junjun He, and Ningsheng Xu. Medq-bench: Evaluating and exploring medical image quality assessment abilities in mllms. *ArXiv*, abs/2510.01691, 2025.
- Kai Liu, Ziqing Zhang, Wenbo Li, Renjing Pei, Fenglong Song, Xiaohong Liu, Linghe Kong, and Yulun Zhang. Dog-iqa: Standard-guided zero-shot mllm for mix-grained image quality assessment. *ArXiv*, abs/2410.02505, 2024.
- Pavan C. Madhusudana, N. Birkbeck, Yilin Wang, Balu Adsumilli, and A. Bovik. Image quality assessment using contrastive learning. *IEEE Transactions on Image Processing*, 31:4149–4161, 2021.
- Zhaoqing Pan, Feng Yuan, Jianjun Lei, Yuming Fang, Xiao Shao, and S. Kwong. Vcnet: Visual compensation restoration network for no-reference image quality assessment. *IEEE Transactions on Image Processing*, 31:1613–1627, 2022.
- N. Ponomarenko, Lina Jin, O. Ieremeiev, V. Lukin, K. Egiazarian, J. Astola, B. Vozel, K. Chehdi, M. Carli, F. Battisti, and C.-C. Jay Kuo. Image database tid2013: Peculiarities, results and perspectives. *Signal Process. Image Commun.*, 30:57–77, 2015.
- Alec Radford, Jong Wook Kim, Chris Hallacy, A. Ramesh, Gabriel Goh, S. Agarwal, G. Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and I. Sutskever. Learning transferable visual models from natural language supervision. *ArXiv*, abs/2103.00020, 2021.
- D. Rymarczyk, Lukasz Struski, Michal G’orszczak, K. Lewandowska, J. Tabor, and Bartosz Zieliński. Interpretable image classification with differentiable prototypes assignment. In *European Conference on Computer Vision*, pp. 351–368, 2021.

- Avinab Saha, Sandeep Mishra, and A. Bovik. Re-iqu: Unsupervised learning for image quality assessment in the wild. *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5846–5855, 2023.
- Shaolin Su, Qingsen Yan, Yu Zhu, Cheng Zhang, Xin Ge, Jinqiu Sun, and Yanning Zhang. Blindly assess image quality in the wild guided by a self-adaptive hyper network. *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3664–3673, 2020.
- Yu Tian, Shiqi Wang, Baoliang Chen, and S. Kwong. Causal representation learning for gan-generated face image quality assessment. *IEEE Transactions on Circuits and Systems for Video Technology*, 34:7589–7600, 2024.
- Yu Tian, Yue Liu, Shiqi Wang, and Sam Kwong. Quality assessment for text-to-image generation: A survey. *IEEE MultiMedia*, 32:44–52, 2025.
- Jianyi Wang, Kelvin C. K. Chan, and Chen Change Loy. Exploring clip for assessing the look and feel of images. In *AAAI Conference on Artificial Intelligence*, pp. 2555–2563, 2022.
- Jiarui Wang, Huiyu Duan, Jing Liu, Shi Chen, Xiongkuo Min, and Guangtao Zhai. Aigciqa2023: A large-scale image quality assessment database for ai generated images. In *CAAI International Conference on Artificial Intelligence (CICAI)*, 2023.
- Haoning Wu, Zicheng Zhang, Erli Zhang, Chaofeng Chen, Liang Liao, Annan Wang, Kaixin Xu, Chunyi Li, Jingwen Hou, Guangtao Zhai, Geng Xue, Wenxiu Sun, Qiong Yan, and Weisi Lin. Q-instruct: Improving low-level visual abilities for multi-modality foundation models. *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 25490–25500, 2023a.
- Haoning Wu, Zicheng Zhang, Weixia Zhang, Chaofeng Chen, Liang Liao, Chunyi Li, Yixuan Gao, Annan Wang, Erli Zhang, Wenxiu Sun, Qiong Yan, Xiongkuo Min, Guangtao Zhai, and Weisi Lin. Q-align: Teaching Imms for visual scoring via discrete text-defined levels. *ArXiv*, abs/2312.17090, 2023b.
- Haoning Wu, Zicheng Zhang, Erli Zhang, Chaofeng Chen, Liang Liao, Annan Wang, Chunyi Li, Wenxiu Sun, Qiong Yan, Guangtao Zhai, and Weisi Lin. Q-bench: A benchmark for general-purpose foundation models on low-level vision. In *International Conference on Learning Representations (ICLR)*, 2024a.
- Tianhe Wu, Kede Ma, Jie-Kai Liang, Yujiu Yang, and Lei Zhang. A comprehensive study of multi-modal large language models for image quality assessment. In *European Conference on Computer Vision*, pp. 143–160, 2024b.
- Tianhe Wu, Jian Zou, Jie-Kai Liang, Lei Zhang, and Kede Ma. Visualquality-r1: Reasoning-induced image quality assessment via reinforcement learning to rank. *ArXiv*, abs/2505.14460, 2025.
- Sidi Yang, Tianhe Wu, Shu Shi, S. Gong, Ming Cao, Jiahao Wang, and Yujiu Yang. Maniqa: Multi-dimension attention network for no-reference image quality assessment. *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pp. 1190–1199, 2022.
- Zhichao Yang, Leida Li, Pengfei Chen, Jinjian Wu, and Giuseppe Valenzise. Language-guided visual perception disentanglement for image quality assessment and conditional image generation. *ArXiv*, abs/2503.02206, 2025.
- Dongshuo Yin, Yiran Yang, Zhechao Wang, Hongfeng Yu, Kaiwen Wei, and Xian Sun. Ipreditions. *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 20116–20126, 2023.
- Zhiyuan You, Zheyuan Li, Jinjin Gu, Zhenfei Yin, Tianfan Xue, and Chao Dong. Depicting beyond scores: Advancing image quality assessment through multi-modal language models. *ArXiv*, abs/2312.08962, 2023.

- Zhiyuan You, Jinjin Gu, Zheyuan Li, Xin Cai, Kaiwen Zhu, Tianfan Xue, and Chao Dong. Enhancing descriptive image quality assessment with a large-scale multi-modal dataset. *IEEE Transactions on Image Processing*, 34:8201–8215, 2024.
- Zhiyuan You, Xin Cai, Jinjin Gu, Tianfan Xue, and Chao Dong. Teaching large language models to regress accurate image quality scores using score distribution. *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 14483–14494, 2025.
- Weixia Zhang, Kede Ma, Jia Yan, Dexiang Deng, and Zhou Wang. Blind image quality assessment using a deep bilinear convolutional neural network. *IEEE Transactions on Circuits and Systems for Video Technology*, 30(1):36–47, 2020.
- Weixia Zhang, Guangtao Zhai, Ying Wei, Xiaokang Yang, and Kede Ma. Blind image quality assessment via vision-language correspondence: A multitask learning perspective. *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 14071–14081, 2023.
- Shijie Zhao, Xuanyu Zhang, Weiqi Li, Junli Li, Li Zhang, Tianfan Xue, and Jian Zhang. Reasoning as representation: Rethinking visual reinforcement learning in image quality assessment. *ArXiv*, abs/2510.11369, 2025.
- Xiaoyu Zhou, Jianwei Fei, Peipeng Yu, Jing Xie, Chong Cheng, and Zhihua Xia. Pgc: Peak-guided calibration for generalizable ai-generated image detection. *ArXiv*, abs/2605.21207, 2026.
- Hancheng Zhu, Leida Li, Jinjian Wu, W. Dong, and Guangming Shi. Metaiqa: Deep meta-learning for no-reference image quality assessment. *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 14131–14140, 2020.

A SUPPLEMENTARY FIGURES AND ZERO-SHOT TRANSFER TABLE

This appendix collects analytical plots that visualize data already tabulated in the main text (Figures 5, 6, 7) together with the zero-shot AIGC / low-level-perception transfer table (Table 6).

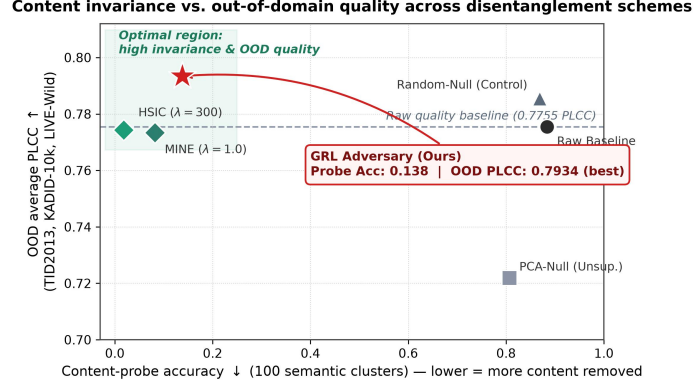


Figure 5: Content-probe accuracy vs. out-of-domain average PLCC across disentanglement schemes (visualizing Table 2). Lower probe accuracy indicates greater semantic content removal. The proposed supervised GRL adversary achieves the optimal trade-off: reducing content probe accuracy to 0.138 while achieving the highest OOD quality PLCC (0.7934).

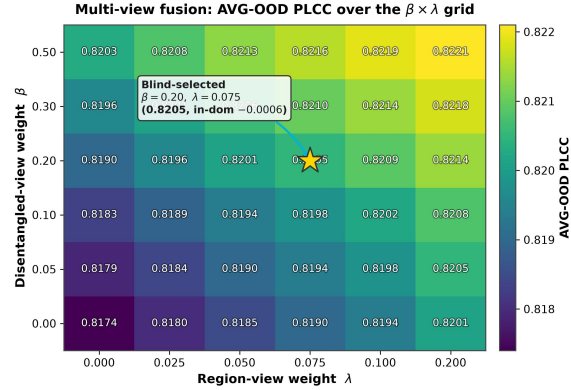


Figure 6: Average OOD PLCC across the multi-view fusion grid ($\beta \times \lambda$), averaged over TID2013, KADID, LIVE-Wild, and SPAQ (visualizing Table 7). The gold star marks the blind-validation selected operating point ($\beta = 0.20, \lambda = 0.075$), which attains 0.8205 AVG-OD PLCC at a negligible in-domain drop (-0.0006 PLCC).

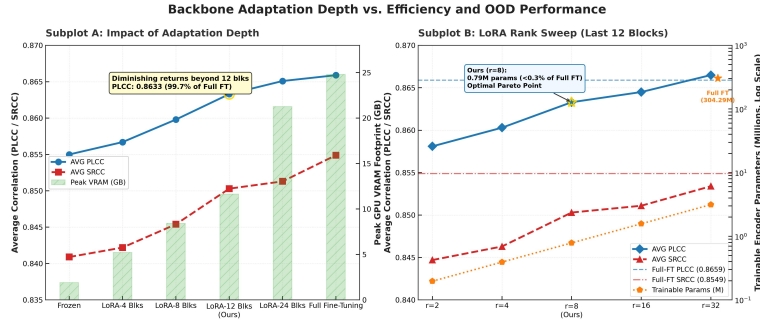


Figure 7: Backbone adaptation depth vs. efficiency and OOD performance (visualizing Table 5). Subplot A displays average correlation metrics and VRAM footprint across adaptation depths. Subplot B illustrates a LoRA rank sweep on the last 12 blocks against trainable parameter count.

Table 6: Zero-shot generalization of the KonIQ-trained model (no fine-tuning or threshold change) to AI-generated content (*AIGC IQA2023*, 2,400 images; PLCC / SRCC) and low-level visual perception (*Q-Bench* / *LLVisionQA* ordinal MCQs; SRCC / Ordinal Acc., random $\approx 35\%$). † Text-image correspondence depends on the unseen prompt and is reported as an honesty control.

Method	AIGC IQA2023 (PLCC / SRCC)			Q-Bench / LLVisionQA (SRCC / Ord. Acc.)	
	Quality	Authenticity	Corresp. †	Overall ($N=137$)	Broad ($N=193$)
RALI Zhao et al. (2025)	0.7964 / 0.7772	0.6091 / 0.6071	0.4823 / 0.4685	0.6484 / 64.96%	0.6289 / 62.69%
DisCoRe-IQA (Ours)	0.8091 / 0.7856	0.6173 / 0.6108	0.4813 / 0.4618	0.6573 / 64.23%	0.6392 / 62.18%

B EXTENDED METHODOLOGICAL DERIVATIONS

B.1 ORTHOGONAL PROJECTION AND GRADIENT REVERSAL DYNAMICS

Consider the quality projection matrix $\mathbf{W}_q \in \mathbb{R}^{M \times M}$ mapping the compressed global visual feature $\hat{\mathbf{z}} \in \mathbb{R}^M$ to the quality space $\mathbf{q} = \mathbf{W}_q \hat{\mathbf{z}}$. The adversarial semantic classifier C_{adv} consists of a linear projection $\mathbf{W}_{\text{adv}} \in \mathbb{R}^{K_{\text{clusters}} \times M}$ followed by a Softmax layer predicting semantic target clusters $y_{\text{content}} \in \{1, \dots, K_{\text{clusters}}\}$. The adversarial cross-entropy loss is formulated as:

$$\mathcal{L}_{\text{CE}}(C_{\text{adv}}(\text{GRL}(\mathbf{q})), y_{\text{content}}) = -\log \frac{\exp(\mathbf{W}_{\text{adv}, y_{\text{content}}}^T \mathbf{q})}{\sum_{k=1}^{K_{\text{clusters}}} \exp(\mathbf{W}_{\text{adv}, k}^T \mathbf{q})} \quad (10)$$

By inserting the Gradient Reversal Layer (GRL) between $\mathbf{q}$ and C_{adv} , the backward gradient propagated to $\mathbf{W}_q$ is reversed:

$$\frac{\partial \mathcal{L}_{\text{dis}}}{\partial \mathbf{W}_q} = \frac{\partial \mathcal{L}_{\text{Huber}}}{\partial y_{\text{dis}}} \frac{\partial y_{\text{dis}}}{\partial \mathbf{q}} \hat{\mathbf{z}}^T - \alpha_{\text{GRL}}(t) \lambda_{\text{adv}} \left(\frac{\partial \mathcal{L}_{\text{CE}}}{\partial C_{\text{adv}}} \mathbf{W}_{\text{adv}} \right) \hat{\mathbf{z}}^T \quad (11)$$

This formulation forces $\mathbf{W}_q$ to project visual embeddings into a subspace orthogonal to the principal components governing semantic cluster separation, thereby maximizing the entropy of C_{adv} and removing category bias.

B.2 METRIC-WEIGHTED PROTOTYPE TRANSPORT FORMULATION

The augmented quality representation $\mathbf{q}_I = [\mathbf{q}; \alpha_{\text{ext}} v_{\text{ext}}] \in \mathbb{R}^{M+1}$ resides on a metric-weighted representation space equipped with learned diagonal metric tensor $\mathbf{W}_{\text{metric}} = \text{diag}(w_1, \dots, w_M, w_{\text{ext}})$. The squared distance to the k -th prototype anchor $\mathbf{a}_k$ is given by:

$$D_k^2(\mathbf{q}_I, \mathbf{a}_k) = \sum_{m=1}^M w_m (q_{I,m} - a_{k,m})^2 + w_{\text{ext}} (\alpha_{\text{ext}} v_{\text{ext}} - a_{k,\text{ext}})^2 \quad (12)$$

Using the dynamic temperature function $\tau(v_{\text{ext}}) = \tau_0 \exp(-\gamma_{\text{temp}} v_{\text{ext}})$, the transport weights $w_k(\tau)$ concentrate probability mass onto boundary prototypes as extreme energy v_{ext} increases:

$$w_k(\tau) = \frac{\exp(-D_k^2(\mathbf{q}_I, \mathbf{a}_k)/\tau(v_{\text{ext}}))}{\sum_{j=1}^K \exp(-D_j^2(\mathbf{q}_I, \mathbf{a}_j)/\tau(v_{\text{ext}}))} \quad (13)$$

Analytical re-calibration for custom MOS scales. Since all benchmarks are normalized to $[1, 5]$, the calibration constants a and b (Eq. 8) act as a fixed linear map; for deployment on a custom range $[S_{\min}, S_{\max}]$ they re-scale analytically as $a' = a(S_{\max} - S_{\min})/4$ and $b' = S_{\max} - (5 - b)(S_{\max} - S_{\min})/4$, without re-training H_{patch} .

C QUALITATIVE VISUALIZATIONS & DEFECT HEATMAPS

Figure 2 illustrates region-level quality segmentation outputs across four representative distortion categories: 1. **Localized Gaussian Blur**: A spatially bounded blur kernel applied to the upper-right corner of a natural landscape image. The attention-entropy saliency map s_n highlights high visual focus, while the patch severity head e_n pinpoints the blurred region ($e_n \approx 0.89$), landing the predicted locus $\hat{n}$ directly inside the ground-truth defect box. 2. **JPEG Blockiness**: Localized compression artifacts on architectural textures. The patch head identifies high high-frequency blocking severity ($e_n \approx 0.78$) with precise bounding box agreement. 3. **Defocus Blur**: Foreground portrait

blurred in the background. The worst-region locus correctly identifies the defocus region without being distracted by sharp facial features. 4. **Sensor Noise:** Localized grain noise on uniform fabric background. Saliency and patch severity heads correctly isolate noisy patches ($e_n \approx 0.84$).

D COMPLETE TABULAR ABLATION CURVES AND SENSITIVITY DATA

Table 7 provides the complete tabular data for the 36-point fusion grid search ($\beta \times \lambda$) evaluated across in-domain protection (*KonIQ-holdout*) and out-of-domain average PLCC/SRCC.

Table 7: Complete 36-Point Multi-View Fusion Grid Search Data ($\beta \times \lambda$).

β (Disentangled)	λ (Region)	KonIQ-Hold PLCC	AVG-OOD PLCC	AVG-OOD SRCC
0.00	0.000	0.9713	0.8174	0.7955
0.00	0.025	0.9713	0.8180	0.7961
0.00	0.050	0.9712	0.8185	0.7966
0.00	0.075	0.9709	0.8190	0.7969
0.00	0.100	0.9704	0.8194	0.7971
0.00	0.200	0.9671	0.8201	0.7971
0.05	0.000	0.9714	0.8179	0.7962
0.05	0.025	0.9714	0.8184	0.7968
0.05	0.050	0.9712	0.8190	0.7972
0.05	0.075	0.9709	0.8194	0.7976
0.05	0.100	0.9705	0.8198	0.7978
0.05	0.200	0.9671	0.8205	0.7977
0.10	0.000	0.9714	0.8183	0.7968
0.10	0.025	0.9714	0.8189	0.7974
0.10	0.050	0.9712	0.8194	0.7979
0.10	0.075	0.9709	0.8198	0.7982
0.10	0.100	0.9705	0.8202	0.7984
0.10	0.200	0.9670	0.8208	0.7984
0.20	0.075 (Ours)	0.9707	0.8205	0.7993
0.20	0.100	0.9703	0.8209	0.7996
0.20	0.200	0.9667	0.8214	0.7995
0.30	0.075	0.9704	0.8210	0.8004
0.30	0.100	0.9699	0.8214	0.8006
0.30	0.200	0.9662	0.8218	0.8004
0.50	0.075	0.9690	0.8216	0.8018
0.50	0.100	0.9685	0.8219	0.8020
0.50	0.200	0.9646	0.8221	0.8017

D.1 ADVERSARIAL LOSS SCHEDULE ANALYSIS (ABLATION 6)

Table 8 reports the full analysis of the gradient-reversal strength schedule $\alpha_{\text{GRL}}(t)$ used to train Module #2, isolating the disentangled view y_{dis} (evaluated alone, without the LoRA base). Starting the adversary at full strength ($\alpha = 1.0$) produces the largest early gradient magnitude and the weakest OOD transfer, confirming that a warm start is required before enforcing content invariance. Gentle or gradually-annealed schedules are consistently more stable; among them the Sigmoid Annealing schedule offers a robust, monotone content-removal trajectory at near-top OOD correlation, which is why it is adopted as the default.

E CALIBRATION AND FUSION-WEIGHT ROBUSTNESS

To confirm that the linear calibration parameters (a, b) and the fusion weights (β, λ) are not over-fit to a specific dataset, we perform two cross-validation studies.

5-Fold Cross-Validation of (a, b) on KonIQ-10k. We fit (a, b) via linear regression ($v_{\text{ext}} \rightarrow \text{MOS}$) on 4 folds of *KonIQ-10k* and evaluate the full pipeline on all 7 test sets using each fold’s fitted values (Table 9). The coefficients are highly stable (coefficient of variation: $a = 0.57\%$, $b = 0.21\%$),

Table 8: Ablation 6: Dynamic adversarial loss schedule $\alpha_{\text{GRL}}(t)$ analysis during training. Metrics are computed for the disentangled view y_{dis} in isolation.

Schedule Strategy	$\ \nabla W_q\ $ Early	Probe Acc ($t = 800$)	KonIQ PLCC	TID2013 PLCC (OOD)
1. Fixed Low ($\alpha = 0.05$)	0.419	0.047	0.9404	0.7514
2. Fixed High ($\alpha = 1.00$)	1.290	0.419	0.9383	0.7302
3. Linear Warmup (t/T)	0.664	0.156	0.9463	0.7401
4. Stepwise ($0 \rightarrow 0.5 \rightarrow 1.0$)	0.337	0.069	0.9470	0.7412
5. Sigmoid Annealing (Ours)	0.763	0.154	0.9459	0.7409

and the resulting AVG-7 benchmark PLCC varies by only $\sigma = 4.88 \times 10^{-6}$, showing that (a, b) generalize across cross-domain distributions.

Table 9: 5-fold cross-validation of the linear calibration parameters (a, b) fit on *KonIQ-10k*.

Fold	a	b	KonIQ Holdout PLCC / MAE	AVG-7 PLCC / MAE
Fold 0	-3.7443	+5.8708	0.9725 / 0.1335	0.8046 / 0.5049
Fold 1	-3.8005	+5.9004	0.9636 / 0.1345	0.8046 / 0.5050
Fold 2	-3.7925	+5.9003	0.9650 / 0.1367	0.8046 / 0.5050
Fold 3	-3.7840	+5.8955	0.9658 / 0.1359	0.8046 / 0.5050
Fold 4	-3.7785	+5.8874	0.9676 / 0.1317	0.8046 / 0.5050
Mean $\pm$ Std	-3.7800 ± 0.0216	$+5.8909 \pm 0.0124$	—	PLCC $\sigma = 4.88 \times 10^{-6}$

Leave-One-Dataset-Out (LODO) Validation of (β, λ) Excluding TID2013. For each held-out OOD test set, (β, λ) are grid-searched on the *other five* OOD datasets (TID2013 excluded entirely; Table 10). The search recovers a consistent regime ($\beta = 0.117 \pm 0.061$, $\lambda = 0.100 \pm 0.000$), and LODO-tuned weights match our fixed paper choice ($\beta = 0.20$, $\lambda = 0.075$) within $\Delta\text{PLCC} \leq 0.0029$ on every test set — consistent with the broad plateau in Figure 6 — so the operating point is stable without relying on TID2013.

Table 10: Leave-One-Dataset-Out validation of fusion weights (β, λ) , excluding TID2013 from the selection process entirely.

Held-Out Dataset	Tuned (β, λ)	Held-Out PLCC / SRCC	Fixed (0.20, 0.075) PLCC / SRCC
SPAQ	(0.15, 0.100)	0.9031 / 0.9034	0.9026 / 0.9030
KADID	(0.00, 0.100)	0.7291 / 0.7311	0.7319 / 0.7356
PIPAL	(0.15, 0.100)	0.5310 / 0.5385	0.5299 / 0.5381
LIVE-Wild	(0.15, 0.100)	0.9068 / 0.8887	0.9071 / 0.8890
AGIQA-3K	(0.10, 0.100)	0.7817 / 0.7142	0.7806 / 0.7138
CSIQ	(0.15, 0.100)	0.8291 / 0.7933	0.8281 / 0.7933
Mean $\pm$ Std	$(0.117 \pm 0.061, 0.100 \pm 0.000)$	—	—

F DETAILED REBUTTAL VALIDATION TABLES

This appendix provides the full numeric tables summarized visually in Figure 4.

Human-gaze validation of s_n (W2/Q2). Table 11 reports the full SALICON 2017 validation metrics ($N=5,000$). The zero-FLOP attention-entropy saliency s_n is upsampled from the final CLIP-ViT layer (24×24 grid at 336^2) and smoothed with a standard $\sigma=14$ px Gaussian; continuous maps and discrete fixations are scored with the standard MIT/SALICON CC, NSS, and AUC-Judd metrics.

Selective disentanglement probe (Q4). Table 12 reports the full linear-probe study contrasting the pre-disentangled embedding $\hat{z}$ with the disentangled quality space q across one high-level semantic target and five low-level quality targets. The signal-retention rate is $\text{score}(q)/\text{score}(\hat{z})$; values near 1.0 indicate the quality signal survives disentanglement, while the near-zero rate on semantics indicates content removal.

Shape generalization and real-world grounding (W4/Q3/S4). Table 13 reports the controlled shape-generalization study of the *frozen* patch head H_{patch} on 500 held-out KonIQ images synthesized with four distinct defect geometries. On the real-world *QGround-100K* test set ($N=1,020$

Table 11: Validation of the zero-FLOP attention-entropy saliency s_n against human gaze on the *SALICON* 2017 validation set ($N=5,000$ images).

Saliency Map	CC $\uparrow$	NSS $\uparrow$	AUC-Judd $\uparrow$
Uniform Spatial Weighting	0.0000	0.0000	0.5000
Center Prior	0.4563	0.7074	0.7003
s_n (Attention-Entropy, Ours)	0.4133	0.7695	0.6819
s_n + Center Prior	0.5469	0.9156	0.7402

Table 12: Selective disentanglement: linear probes on the pre-disentangled space $\hat{z}$ vs. the disentangled quality space $q = W_q \hat{z}$. High-level semantics are stripped; low-level quality is preserved.

Evaluation Target	Pre-Disent. $\hat{z}$	Disent. q	Retention
High-Level Semantics (100 CLIP clusters, Acc)	0.855	0.122	0.14 (<i>stripped</i>)
Low-Level Quality (KonIQ MOS SRCC)	0.900	0.917	1.02
Low-Level Quality (KADID MOS SRCC)	0.921	0.784	0.85
Quality Severity Quantiles (KonIQ 2-bin Acc)	0.847	0.885	1.05
Quality Severity Quantiles (KonIQ 5-bin Acc)	0.533	0.589	1.10
Within-Distortion Severity Rank (KADID SRCC)	0.787	0.714	0.91

human-annotated regions), H_{patch} attains a region bounding-box IoU of 0.6703 (IoU@0.5 = 0.6852), matching the grounding performance in Figure 3(B); its point hit-rate there (56.8%) reflects the benchmark’s frequent partial masks on globally degraded images, as classical local blur/texture descriptors likewise hover near chance on the same set (AUC 0.414–0.534).

Table 13: Controlled shape-generalization of the frozen patch head H_{patch} (trained only on rectangular boxes) on 500 held-out images across four defect geometries.

Mask Shape	Hit-Rate ($\arg \max e_n$) $\uparrow$	Per-Image ROC-AUC $\uparrow$	Mean BBox IoU $\uparrow$
Rectangle (Training Control)	0.934	0.825	0.511
Rotated Ellipse	0.854	0.791	0.518
Irregular Polygon	0.851	0.790	0.518
Soft Organic Blob	0.875	0.757	0.682
Organic Average	0.860	0.779	0.573